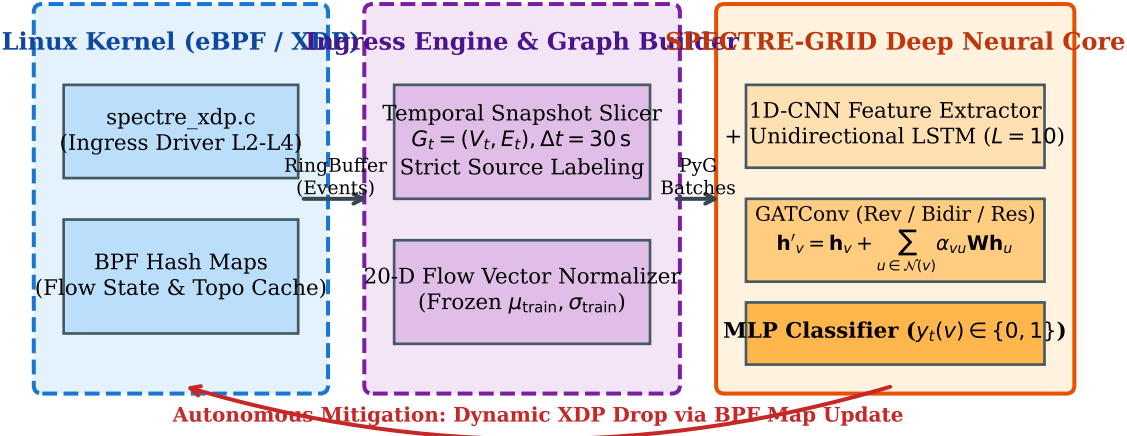
Linux Kernel (eBPF / XDP)
spectre_xdp.c
(Ingress Driver L2-L4)
BPF Hash Maps
(Flow State & Topo Cache)
RingBuffer
(Events)
Ingress Engine & Graph Builder
Temporal Snapshot Slicer
$G_t = (V_t, E_t), \Delta t = 30\,\mathrm{s}$
Strict Source Labeling
20-D Flow Vector Normalizer
(Frozen $\mu_{\mathrm{train}}, \sigma_{\mathrm{train}}$)
PyG
Batches
SPECTRE-GRID Deep Neural Core
1D-CNN Feature Extractor
+ Unidirectional LSTM ($L = 10$)
GATConv (Rev / Bidir / Res)
$\mathbf{h}'_v = \mathbf{h}_v + \sum_{u \in \mathcal{N}(v)} \alpha_{vu} \mathbf{W} \mathbf{h}_u$
MLP Classifier ($y_t(v) \in \{0, 1\}$)
Autonomous Mitigation: Dynamic XDP Drop via BPF Map Update